MIL-P-60622B (AR)
14 September 1977
SUPERSEDING
MIL-P-60622A (MU)
1 November 1971

MILITARY SPECIFICATION

PROJECTILE, 175MM: HE, M437
METAL PARTS ASSEMBLY

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the manufacturing and assembly of metal part for Projectile, 175MM: HE, M437.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.-The following documents of the issue in effect on date of invitation for bids, form a part of this military specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For.
MIL-A-2550 - Ammunition, General Specification For

STANDARDS

FEDERAL

Federal Test Method Standard No. 141 - Paint, Varnish, Lacquer and Related Materials; Method of Inspection and Testing

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-410 - Nondestructive Testing Personnel Qualifications and Certification
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS (see 6.12)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

10520193 - Projectile, HE, 175MM, M437 Metal Parts Assembly
8857344 - Pallet for Projectile, 175MM
8596059 - Indentor
9280262 - Ultrasonic Inspection Standard Circumferential Cracks & Piping, M437 Projectile
9280263 - Ultrasonic Inspection Standard Longitudinal and Radial Cracks M437 Body
9280264 - Envelope Drawing Ultrasonic Inspection Equipment for M437 Projectile

PUBLICATIONS

FRANKFORD ARSENAL

Operating Instructions for Use of Detroit Testing Machine for Band Heating Inspection

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

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American Society for Testing and Materials (ASTM)

A711 - Carbon Steel Blooms, Billets and Slabs for Forging.
E8 - Tension Testing of Metallic Materials
E10 - Brinell Hardness of Metallic Materials

(Copies may be purchased directly from the American Society for Testing & Materials, 1916 Race Street, Philadelphia, Pa. 19103).

ANSI B46.1 - Surface Texture
ANSI Y32.3 - Welding Symbols
ANSI Y14.5 - Dimensioning and Tolerancing

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y., 10018).

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Projectile assembly.-The projectile assembly and components shall comply with all requirements specified on Drawing (Dwg.) 10520193 and with all requirements specified in applicable specifications.

3.3 Steel quality.-The item contractor shall establish procedures to provide a Forge Quality Steel from ASTM A711, i.e., the steel shall be free of pipe, porosity and heavy segregation.

3.4 Projectile body.

3.4.1 Hot Forge - Heat treat method of manufacture.-Each projectile body manufactured by this method shall be subjected to heat treatment to attain required mechanical properties specified on the applicable drawing.

3.5 Mechanical properties.

3.5.1 Body.-Minimum mechanical properties as required by the applicable drawing shall apply to specimens machined from any part of the projectile body.

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3.5.2 Rotating band.-After application and subsequent removal, the rotating band shall have the following mechanical properties:

Tensile Strength	- 37,000 lbs. per square inch (PSI) minimum
Elongation	- 25 percent. minimum in one inch gage length

3.6 Banding.

3.6.1 Band seat.-Prior to banding, the band seat shall be clean and free from oil, grease, dirt, rust or other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seat after knurling.

3.6.2 Application.-The rotating band shall be applied after final heat treatment of the projectile body, by carefully pressing the band blank into the band seat so as to make contact with the sidewall to a depth not less than .125 inch measured from the bourrelet surface in a direction normal to the bourrelet surface. There shall be no gap between the seated blank and the sidewall at the bourrelet surface. The band shall be applied in such a manner as not to distort the projectile body. Distortion shall be considered not to have taken place if the projectile body immediately adjacent to the band seat is within specified drawing tolerances.

3.6.3 Band seating.-The diametrical clearance between the rotating band and band seat shall not exceed .012 inch.

3.7 Proving ground.-The assembled projectile metal parts shall reveal no evidence of failure when tested at excess and service pressures. Metal parts failure shall be:

1. Projectile body separation attributable to noncompliance with drawing and specification requirements.
2. Loss of metal parts components.
3. Loose rotating band attributable is noncompliance with applicable drawing and specification requirements.
4. Shear of rotating band as evidenced by complete lack of engraving.

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3.8 Ultrasonic inspection.-The projectiles shall be free of any discontinuities equal to or in excess of the inspection criteria within their respective zones. The inspection criteria are notches and holes in standards (Per Drawing Numbers 9280262 and 9280263) which simulate cracks and pipes.

3.9 Projectile metal defects.

3.9.1 Projectile body.-The body, including the cavity, shall be free from cracks, splits, burst, cold shuts, pipe, porosity, inclusions, folds, seams, and other metal defects.

3.9.2 Projectile body cavity.-The body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.9.3 Visual standards.-Visual standards for evaluation of metal defects will be established for each contract after the first months production. However, the establishment of visual standards will not waive the requirement for compliance with 3.9.1 and 3.9.2 (see 6.11).

3.10 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

3.11 Interior cavity finish.

3.11.1 The interior cavity after painting shall be free from bare spots, grease, dirt, or other foreign matter.

3.11.2 There shall be no blisters or pool of paint in the bottom of the cavity prior to palletizing.

3.12 Surface finish.-The requirements for surface finish are as detailed on the applicable drawing. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

3.13 Workmanship.-The requirements for workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

3.13.1 Surface preparation.-Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced thereon. Where cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

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3.13.2 Painting.-Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.13.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing, and shall be in accordance with the provisions of specification MIL-A-2550.

3.13.4 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.13.5 Foreign matter.-No part of assembly shall contain dirt, grease, chips, corrosion, or other foreign matter. Particular attention shall be paid to the cavity of the body assembly, prior to painting, to insure that no grit, scale or other foreign matter remains therein after final cleaning.

3.14 Base cover weld.-The base cover shall be welded to the base and the welding operation shall be so controlled as to assure fusion between the projectile and base cover. The welding cycle shall include appropriate stress relief. ANSI Y32.3 shall be used to interpret welding symbols and definitions.

3.15 Tool control dimensions.-Dimensions marked tool control shall be gaged at the beginning of production and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawings	Quantity
Projectile,HE,175MM,M437 Metal Parts Assembly	10520193	45
Body Assembly (unpainted)	10520194	2
Body Assembly (Prior to	10520194	2 per machine

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60622B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile Assembly and Components	1 OF 1	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, HE, 175MM, M437 Metal Parts Assembly (Dwg. 10520193)				
	Examination for Defects	45		3.2	4.4.2.5
	Proving Ground Test	20		3.7	4.5.7
	Body Assembly (unpainted) (Dwg. 10520194)				
	Examination for Defects	2		3.2	4.4.2.4
	Mechanical Properties Test	2		3.5	4.4.3.2.2
	Visual Inspection	2		3.9.3	4.5.2
	Body Assembly (Prior to cutting threads and Band seat (Dwg. 10520194)				
	Ultrasonic Inspection	45		3.8	4.5.3
	Body Assembly (Prior to machining band) (Dwg. 10520194				
	Rotating Band Tension Test	2 per machine		3.5.2	4.4.3.2.2.1

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

In addition the heat treatment inspection lots shall be restricted to components heat treated in the same furnace at the same temperature and for the same length of time. Test samples shall be selected after final heat treatment. Where heat treatment is performed at the same time in a batch-type furnace, additional identification control must be maintained which directly relates to that portion of the inspection lot processed as a batch. The method of control must meet with approval of the Government. Components in heat treatment inspection lots shall remain grouped together for identifying purposes until mechanical property tests and interior surface and internal radius examinations have been completed. Components from heat treatment inspection lots which comply with applicable requirements may be subsequently regrouped into final assembly lots.

4.2.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60622B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
44.2.1	Projectile Body Before Nosing		10520195
			NEXT HIGHER ASSEMBLY 10520193

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Defective cavity		100%	3.9.2	Visual with probe light
Major					
101	Length from base of cavity to rear basic diameter of cavity		0.40%	3.2	Gage
102	Length from base of cavity to forward basic diameter of cavity		0.40%	3.2	Gage
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60622B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Body Assembly (Prior to cutting band seat and threads)				10520194
					NEXT HIGHER ASSEMBLY 10520193
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Ultrasonic test		100%	3.8	4.5.3
2	Ultrasonic Test Stamp missing		100%	3.8	Visual
Major	None defined				
Minor	None defined				

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60622B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Projectile Body (After Nosing)		10520195
			NEXT HIGHER ASSEMBLY
			10520193

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.9.1	Visual with probe light
Major					
101	Diameter of rotating band seat		0.40%	3.2	Gage
102	Thickness through the base		0.40%	3.2	Gage
103	Scoring missing from rotating band seat		0.40%	3.2	Visual
104	Foreign matter on rotating band seat		0.40%	3.6	Visual
Minor					
201	Location of rotating band seat		0.65%	3.2	Gage
202	Length total		0.65%	3.2	Gage
203	Width of rotating band seat		0.65%	3.6	Gage
204	Evidence of shot or sandblasting of the rotating band seat		0.65%	3.6	Visual
205	Radii of chamfers missing or incorrect		0.65%	3.2	Visual
206	Surface finish improper		0.65%	3.12	Visual
207	Foreign matter (except as otherwise classified)		0.65%	3.13	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.42.4	Body Assembly (Prior to Paint)		10520194
			NEXT HIGHER ASSEMBLY 10520193

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical I Major	Metal defective		100%	3.2/3.9.2	Visual/Probe Light
101	Weight		0.40%	3.2	Approved Scale
102	Concentricity of pitch diameter of threads with bourrelets, including perpendicularity of front face of nose		0.40%	3.2	Gage
103	Concentricity of ogive with bourrelets		0.40%	3.2	Gage
104	Concentricity of body with bourrelets		0.40%	3.2	Gage
105	Concentricity of rotating band ribs wi bourrelets		0.40%	3.2	Gage
106	Concentricity of rotating band lip with bourrelets		0.40%	3.2	Gage
107	Concentricity of rotating boatail at basic diameter with bourrelets		0.40%	3.2	Gage
108	Diameter of bourrelets, min.		0.40%	3.2	Gage
109	Diameter of body		0.40%	3.2	Gage
110	Diameter of rotating band ribs		0.40%	3.2	Gage
111	Diameter of rotating band rib grooves		0.40%	3.2	Gage
112	Diameter of rotating band lip		0.40%	3.2	Gage
113	Diameter of rotating band rear of lip		0.40%	3.2	Gage
114	Diameter of the obturator seat		0.40%	3.2	Gage

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PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 10520194
4.4.2.4	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY 10520193
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
115	Length of body undercut		0.40%	3.2	Gage
116	Location of forward outside basic diameter		0.40%	3.2	Gage
117	Location of forward boattail basic diameter from base		0.40%	3.2	Gage
118	Location of rear boattail basic diameter from base		0.40%	3.2	Gage
119	Length from front face to rear of forward bourrelet		0.40%	3.2	Gage
120	Thickness of wall at ogive, min.		0.40%	3.2	Gage
121	Open band		0.40%	3.2	Visual
122	Variation in wall thickness		0.40%	3.2	Gage
123	Variation in wall thickness of ogive		0.40%	3.2	Gage
124	Foreign matter in charge cavity		0.40%	3.13.5	Visual
Minor					
201	Diameter of nose		0.65%	3.2	Gage
202	Depth of undercut forward and rear of rotating band (if applicable)		0.65%	3.2	Gage
203	Location of rotating band		0.65%	3.2	Gage
204	Location of obturator band seat		0.65%	3.2	Gage
205	Length from front of rotating band to front of rotating band lip		0.65%	3.2	Gage
206	Length of angle at front of rotating band		0.65%	3.2	Gage

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
4.4.2.4	CONTINUATION SHEET		10520194
			NEXT HIGHER ASSEMBLY
			10520193

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
207	Width of undercuts forward and rear of rotating band (if applicable)		0.65%	3.2	Gage
208	Width of rotating band		0.65%	3.2	Gage
209	Width of obturator band seat		0.65%	3.2	Gage
210	Surface finish, improper		0.65%	3.2	Gage
211	Radii or chamfer missing or incorrect		0.65%	3.2	Visual
212	Foreign matter (unless otherwise classified)		0.65%	3.13.5	Visual
213	Evidence of poor workmanship		1.00%	3.13	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.5	Metal Parts Assembly				10520193
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Pool of paint in cavity		100%	3.11.2	Visual/Manual
2	Metal Defective		100%	3.9.2	Visual with probe light
Major					
101	Pitch diameter of thread, max.		0.40%	3.2	Gage
102	Minor diameter of thread, max.		0.40%	3.2	Gage
103	Diameter of bourrelets, max.		0.40%	3.2	Gage
104	Length of thread stock, max.		0.40%	3.2	Gage
105	Length of effective thread, min.		0.40%	3.2	Gage
106	Rotating band damaged		0.40%	3.13	Visual
107	Pitch diameter of thread, min.		0.40%	3.2	Gage
108	Minor diameter of thread, min.		0.40%	3.2	Gage
Minor					
201	Paint not dry		0.65%	3.13.2	Visual/Tactile
202	Paint on threads		0.65%	3.2	Visual
203	Paint on rotating band bearing surfaces		0.65%	3.2	Visual
204	Interior protective coating inadequate		0.65%	3.11	Visual
205	Exterior protective coating inadequate or damaged		0.65%	3.2	Visual
206	Corrosion preventive compound missing from threads		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60622B (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.5	CONTINUATION SHEET		10520193
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
207	Lifting plug improperly seated		0.65%	3.2	Visual/Manual
208	Grommet insecure		0.65%	3.2	Visual/Manual
209	Marking missing, incorrect or illegible		0.65%	3.2	Visual
210	Evidence of poor workmanship		1.00%	3.13	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Pallet		8857344
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Pallet damaged		0.65%	3.2	Visual
202	Pallet improperly assembled		0.65%	3.2	Visual
203	Strapping missing, incorrect or illegible		0.65%	3.2	Manual/Visual
204	Marking missing, incorrect or illegible		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.00%	3.13	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60622B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Cover Base		10520198
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.1	Gage
Minor					
201	Diameter		0.65%	3.1	Gage

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4.4.3 Testing

4.4.3.1 Steel quality.-Records of test documentation shall be maintained and made available for Government review to provide evidence of compliance with the requirements of 3.2.

4.4.3.2.1 Hardness.-In order to determine uniformity of heat treatment, each body forging in the heat treatment lot shall be subjected to a hardness test with a Brinell Hardness Tester or any other method approved by the procuring activity.

4.4.3.2.2 Tension test.-Two projectile bodies, one of the hardest and one of the softest as determined by 4.4.3.2.1 shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot.

4.4.3.2.2.1 Rotating band.-Two body assemblies shall be selected from each 8-hour production or fraction thereof for each banding machine in operation. Bands removed from these body assemblies shall be used in mechanical property tests. Failure of any band to comply with requirements of 3.5.2 shall be cause for rejection of those assemblies represented by the sample.

4.4.3.2.3 Base cover weld.-Five projectile bodies (after welding the base cover) from each hours production or fraction thereof per welding machine in operation shall be subjected to this test. Failure of any weld to comply with 3.14 shall be cause for rejection of the production represented by the sample.

4.4.3.2.4 Rotating band tightness.

4.4.3.2.4.1 Nondestructive method.-Five body assemblies from each 1-hour production or fraction thereof per banding machine in operation shall be subjected to this test. Failure of any band to comply with requirements of Indentor Drawing 8593429 shall be cause for rejection of the assemblies represented by the sample.

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4.4.3.2.4.2 Destructive method.-When the equipment for the nondestructing method of testing is not available, one body assembly shall be selected from each 8-hour production or fraction thereof per banding machine in operation. Failure of any band to comply with 3.6.3 shall be cause for rejection of the production represented by the sample.

4.4.3.2.4.3 Depth of contact.-Samples supplied in accordance with 4.4.3.2.2.1 shall be examined to determine depth of contact between band blank and sidewall of the band seat. Failure to comply with the requirements of 3.6.2 shall be cause for rejection of those assemblies represented by the sample.

4.4.3.2.5 Protective coating.-The sample size and acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action prescribed therein.

4.4.3.3 Proving ground.-Testing shall be performed as specified in 4.5.7.

4.4.3.3.1 First article sample.

4.4.3.3.1.1 Initial production sample.-The quantities listed in Table II shall be ballistically tested in the phases indicated.

TABLE II

Test Phase	Quantity
Excess pressure	10
Service pressure	10

4.4.3.3.2 Production lot sample.-The quantities listed in Table III shall be ballistically tested in the phases indicated.

TABLE III

Test Phase	Quantity
Excess pressure	10
Service pressure	10

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4.4.3.3.3 Excess pressure.-Samples provided in accordance with Tables II and III will be subjected to this test. Failure of any projectile assembly to comply with 3.7 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.4.3.3.4 Service pressure.-Samples provided in accordance with Tables II and III will be subjected to this test. Failure of any projectile assembly to comply with 3.7 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.4.3.3.5 Test validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be considered invalid and a new test shall be performed with additional samples.

4.4.4 Inspection equipment.-The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.5 Test methods and procedures

4.5.1 Protective coating.-The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specification listed thereon. The salt spray test shall be performed utilizing equipment prescribed by Federal Test Method Standard No. 141.

4.5.1.1 Salt Spray Testing.-Salt spray testing shall be accomplished on production projectiles or sectioned projectiles that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned projectiles are used, sectioning shall be accomplished to allow testing of all major areas of the projectile on an alternating basis.

4.5.2 Visual examination.-The internal cavity of the projectile shall be inspected visually prior to nosing using peripheral lighting to assure conformance with the applicable requirements. Visual standards in accordance with 3.9.3 shall be set up for use during this inspection. Any projectile failing to meet the applicable requirements of 3.9 shall be classed defective and removed from the lot.

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4.5.3 Ultrasonic inspection.-The projectile shall be ultrasonically inspected prior to cutting band seat and threads using equipment and procedures in accordance with 4.5.3.1. The projectile shall be inspected with ultrasonics throughout all of the material. Ultrasonic energy shall be used at right angles to assure that orientation does not prevent detection of a defect. Standards in accordance with 4.5.3.3 shall be used to check the effectiveness of the inspection process. Any projectile failing to meet the requirements of 3.8 shall be classed defective and removed from the lot.

4.5.3.1 Equipment and procedures.-The ultrasonic inspection shall be performed with the use of a liquid to assure a reliable coupling to transmit ultrasonic energy. Should a stream or thin film of liquid be used to couple this energy, controls shall be used to monitor the coupling and assure its effectiveness. Such monitoring is not required if submergence in a liquid is used.

Using ultrasonic energy at right angles requires at least two separate sources of ultrasonic energy. This means that in the nose, ogive and bourrelet regions, ultrasonic energy shall move within the walls of the projectile with separate significant axial and circumferential components of velocity to detect flaws located anywhere on the outside, inside or within the walls. For the base region, significant ultrasonic velocity components shall separately be radial and circumferential.

Criteria for detection of cracks are grooves which simulate actual cracks. Details are contained on Drawings 9280262, 9280263 and 9280264. Criteria for detection of piping in the base region consists of drilled holes oriented parallel to the major axis of the projectile.

All grooves and holes machined on the standards are targets for rejection. That is, any projectile that sends back an echo signal that is greater than or equal to the signal from the standard shall be rejected.

4.5.3.2 Calibration and operation procedures and equipment.-Prior to initiation of ultrasonic inspection, the contractor shall submit for written approval one copy of his proposed procedures, including descriptions of equipment (contractor design) and operation procedures (see 6.3). No change to the procedure shall be made after the original approval without submission and approval of the proposed changes. As a minimum the adequacy of the inspection process

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shall be checked by using the standards at the start and finish of each production shift or change in operator, as well as at four (4) hour intervals during production. All channels shall properly indicate a rejection signal when encountering the notches and holes mentioned in 3.8. If the inspection process is determined to be inadequate, the condition shall be corrected and all items inspected since the last acceptable inspection check shall be reinspected at the discretion of the government inspector. Equipment used shall provide for a positive activation of a buzzer or alarm as well as a light system for each channel to alert the operator of the signal indications that are the criteria for rejection. A log book shall be maintained by the operator of each system which as a minimum, documents the use of standards, number of items inspected, number of rejects by zone, changing of transducers, instrumentation, repairs, reasons for equipment breakage, if any, and changing of qualified operators or inspectors. This log book shall be made available when requested for Government review.

4.5.3.3 Ultrasonic standards.-Ultrasonic inspection standards shall be provided by the contractor for use during the ultrasonic inspection. These standards shall be machined from acceptable projectiles in accordance with drawings 9280262, 9280263, and 9280264 and shall be inspected for all drawing requirements to determine their acceptability for use. Government verification of all inspections performed on the standards is required. In addition, a certificate of conformance for the standards shall be submitted to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey, 07801, prior to use of the standards.

4.5.3.4 Operator qualification.-Personnel performing the ultrasonic inspection shall be certified in accordance with MIL-STD-410 with the following exception:

Each facility shall have as a minimum an individual who qualifies as a Level II inspector in all respects but need not include experience.

4.5.4 Mechanical properties - Hardness.-After final heat treatment and scale removal, the hardness test shall be performed in the area of forward bourrelet. The test shall be performed using procedures as prescribed by ASTM E-10.

4.5.4.1 Tension test.-From the projectile selected in accordance with 4.4.3.2.2, two tension test specimens will be selected and tested in accordance with ASTM E-8 and these specimens shall be taken longitudinally, from opposite sides of the body. (Midwall of the band seat area). The number of

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specimens and their location are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties of the projectile meet the drawing requirements. The criteria for performing the tests shall be as prescribed by ASTM E-8. Yield strength shall be determined by the offset method. The limiting set shall be 0.1 percent.

4.5.4.1.2 Rotating band.-Following the examination for depth of sidewall contact as described in 4.5.6.3 the rotating band shall be removed from one of the two segments of each sample provided in accordance with 4.4.3.2.2.1 and carefully flattened without hammering to minimize additional cold working. The specimen shall be machined so that all serration marks from the band seat are removed. A test specimen shall conform to ASTM E-8 with a gage length of one inch.

4.5.5 Base cover weld.-The test of the base cover weld shall be performed as follows: Visually inspect the base cover and reject the area that appears to have the weakest weld; insert a 5/8" chisel in this area and apply force by use of a 1-pound steel hammer and an approximately 1-foot stroke. If the weld breaks loose after two strokes, the body assembly will be classed defective. If the weld is found to be secure, the burr made by the chisel may be knocked back into position and accepted.

4.5.6 Rotating band tightr ss.

4.5.6.1 Nondestructive method.-The samples provided in accordance with 4.4.3.2.4.1 shall be tested utilizing the equipment and procedures prescribed in Frankford Arsenal Detroit Testing Machine Manual and Indentor Drawing number 8593429. Two readings, 90 degrees apart at the front edge of the rotating band and two readings 90 degrees apart at the rear edge of the rotating band shall be taken to determine compliance with the established requirements for band tightness depicted on Drawing 8593429.

4.5.6.2 Destructive method.-The sample provided in accordance with 4.4.3.2.4.2, prior to measuring, shall be given a light surface cut in order to true the band and permit accurate measuring. Prior to removal of the band, the diameter of the finished band shall be measured on three diameters 60 degrees apart, and on two points along the cylindrical portions of its length (at the front and at the rear edge). This comprises six diametrical measurements. These points of measurement or orientation points shall be marked on the bands and on the projectile body above and below the bands in such manner that each measurement location can be identified.

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Care shall be exercised in the removal of the bands so as to keep deformation to a minimum and assure the markings will not be obliterated. After removal of the band, the thickness of the band shall be measured at each of the marked or orientation points (a total of 12 thickness measurements on each band); the diameter of the band seat at corresponding points shall also be measured. To each of the measurements of the diameters of the band seat the sum of the corresponding band thickness shall be added and the results subtracted from the corresponding exterior diameters of the bands will indicate the diametrical band gap.

4.5.6.3 Depth of contact.-The samples provided under 4.4.3.2.2.1 shall be given a light surface cut to remove the indent impressions of the banding machine and then sectioned by cutting through two planes normal to the longitudinal axis 1/2 inch forward and rear of the rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two parts shall be used for establishment of mechanical properties of the band (4.5.4.1.2); the other shall be sectioned into two parts thus exposing four surfaces, each of which reveals a view of the band as swaged into the band seat. The exposed surfaces shall be subjected to a light grinding operation to remove metal smear effects of saw cuts prior to measurement for depth of sidewall contact to determine compliance with requirements.

4.5.7 Proving ground functional tests.

4.5.7.1 Excess pressure.-The metal parts assemblies supplied in accordance with Tables II and III, inert loaded, and dummy fuzed, will be fired for recovery in a gun for which the projectile is standard, the tube of which will have 50 to 25 percent life remaining based on current gun tube condemnation criteria. A propellant charge weight adjusted to produce 112 percent of the service pressure of the projectile in a new gun shall be used. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile. Each recovered projectile will be examined for the characteristics listed in 3.7.

4.5.7.2 Service pressure.-The metal parts assemblies supplied in accordance with Tables II and III will be loaded, fuzed and fired using charge III, unless otherwise specified. The projectile assemblies will be fired alternately with projectile assemblies from a reference lot for comparison. The projectile shall be fired in a gun for which the round is standard, the tube of which shall have a minimum remaining

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service life of 50 percent based on current gun tube condemnation criteria. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile. Range velocity and chamber pressure for each round fired will be recorded.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Drawing 8857344.

5.1.1 Level B.-Not applicable (see 6.6).

5.1.2 Level B-1.-Projectile metal parts assemblies shall be palletized in accordance with Drawing 8857344.

6. NOTES

6.1 Intended use.-The components covered by this specification are intended for use on 175MM, M437, HE Projectile.

6.2 Ordering data.-See MIL-A-48078 (PA).

6.3 Submission of inspection procedures and/or equipment designs for approval - See MIL-A-48078.-Submit equipment designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey, 07801.

6.4 It is the responsibility of the manufacturer of the projectile metal parts to select a fully-killed carbon steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of the forger and machiner to produce the projectile to the dimensions and properties specified as an end item requirement.

6.5 Visual examination qualification.-When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.6 The requirements herein for Level B are equivalent to the requirements for full Level B use in accordance with AR 700-15, 20 June 1975. The requirements under Level B-1 are intended to provide for limited storage life necessary to meet on-going loading, assembly and packing operations and to provide uniform packing from all procedures of an item to all using facilities.

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6.7 Submission of ballistic test data and ammunition data cards.-In addition to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data and ammunition data cards shall be forwarded to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-Q and DRDAR-LCU-D, Dover, New Jersey, 07801.

6.8 The supplier of the selected steel raw material is responsbile only for furnishing a metal composition, grade, and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations identifiable with the standard AISI grade selected by the projectile manufacturer and complying with specification ASTM Designation A711, with the exception of supplemental sections SI-S16 inclusive.

6.9 In the application of carbon steel, blooms, billets, and slabs, forging quality processing techniques adapted to use of the selected steel should be applied with the expectation that there will be variables in the metal crease-sectional dimensions in weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized of the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment, sequence of operations, tooling designs, handling devices, and processing techniques having capabilities of compensating for such variables.

6.10 Heat treatment is understood to be any thermal process which brings about a modification in the mechanical or metallurgical properties of the metal involved.

6.11 Visual standards.-Visual standards to be established at contractors plant in compliance with 3.9.3.

6.12 Drawings.-Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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